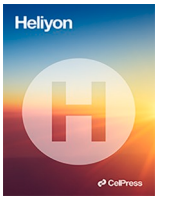
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Research article

Performance comparison between PID and Fuzzy logic controllers for the hardware implementation of traditional high voltage DC-DC boost converter

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ABSTRACT

This research introduces a hardware implementation of DC-DC boost converter designed to elevate the DC voltage generated by renewable sources while effectively regulating it against line and load fluctuations for inverter application. The main objective is to boost the DC link voltage to the level of V_{max} in the output AC voltage obtained from inverter circuits. This enables the inverters for transformer-less power conversion from DC to AC to reduce magnetic losses, size and weight of the inverter circuits used in the utility application. The proposed converter's topology and switching sequences play a crucial role in enhancing overall performance. Utilizing a Zero Current Switching (ZCS) technique, the converter efficiently recovers stored energy from the magnetics. The proposed converter attained the output voltage of 350 V at its current of 1A from the input voltage of 20 V at its current of 19 A. The ZCS technique and the topology of the converter enhances the efficiency to 92 %. The study employs traditional Proportional-Integral (PI) and Proportional-Integral-Derivative (PID) controllers for effective voltage regulation, analysing time domain specifications. Additionally, a Fuzzy logic controller is introduced as an alternative to PID controllers to compare their performance metrics, evaluating the optimization of the converter's transient and steady-state behaviours. The proposed converter is designed, simulated and their performance metrics are analysed using MATLAB for both with and without controllers. The step-time characteristics of the proposed converter with load resistance of R_L = 500 Ω and an input voltage of V_i = 20 V has been determined and analysed. The PID system attained a rise time of 88.781 ms, an overshoot value of 9.341 %, and a steady-state error of 0.00043. The fuzzy system achieved a low-rise time of 10.624 ms, a low overshoot of 0.55 %, and a steady-state error of 0.0584. The hardware prototype of the proposed converter is implemented with a FPGA based PID and Fuzzy logic controllers for providing better voltage regulation and to improve the performance metrics of the converter. The simulation and experimental findings are contrasted, examined, and confirmed to ensure improved consistency in performance measures.

1. Introduction

DC-DC converters commonly accomplish the transformation of DC voltage from one level to another. DC-DC converters are widely used to increase the DC voltage in electric cars [1,2], and [3]. The DC-DC boost converter is commonly referred to as a step-up

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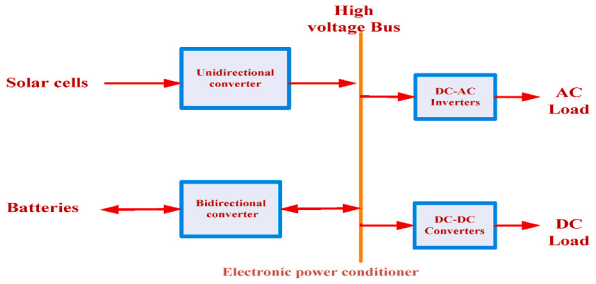


Fig. 1. Architecture of power distribution system [12].

converter when used for high-voltage applications [4]. The literature analysis emphasizes the use of Fuzzy logic controllers to tackle issues caused by non-linear processes. These controllers entail complex calculations and do not require accurate information or a mathematical model [5]. The strong and resilient characteristics of Fuzzy logic controllers enable the incorporation of new rules, making it easy to adapt [6]. To ensure a consistent output, it is necessary to combine the Fuzzy logic controller with the converter in order to satisfy the desired voltage parameters. The regulation of the output of the DC-DC converter is often accomplished by controlling the duration of the ON and OFF states and adjusting the duty ratio of the switching component. The regulatory process is influenced by both linearly operated controllers, such as proportional-integral (PI) and proportional-integral-derivative (PID), as well as non-linear controllers. The Fuzzy logic controller represents the non-linear controllers [7]. Fuzzy logic utilizes expert knowledge to convert linguistic control techniques into automated control systems [8]. Although PID controllers are easy to design, they may not be sufficient for properly handling Fuzzy logic phenomena. Nevertheless, the Fuzzy logic controller encounters a tuning issue, which is regarded as its primary limitation [9].

A novel design for a Fuzzy-logic controller for the buck-boost converter was suggested, employing an exponential version of the PID controller with multiple loops and an extra tuning parameter [10]. The design of a Fuzzy logic controller is dependent on the converter's heuristic knowledge, whereas the design of PID and PI controllers is grounded in the converter's frequency response [11]. Boost converters employing Fuzzy logic controllers provide greater stability compared to those utilizing PI and PID controllers, in both transient and steady-state conditions [9].

Fig. 1 illustrates the structure of a power distribution system. The solar cells are supplied with a unidirectional converter, whereas the batteries are supplied with a bidirectional converter. The DC-AC inverters supply power to the AC load, whereas the DC-DC converters supply power to the DC load. The converter based on Zero Current Switching (ZCS) attained an efficiency of 93.33 % when operated at a frequency of 1 kHz. It was able to generate an output voltage of 350 V from an input voltage of 20 V [13]. The high-capacity converter provided a power output of 480 W when supplied with an input voltage ranging from 39 to 55 V, and produced an output voltage of 48 V. The converter attained a 97 % efficiency at a frequency of 100 kHz [14]. The converter achieved an output of 240 V with a precise efficiency when given an input of 400 V. The converter had a remarkable efficiency of 99.9 % at a frequency of 23 kHz [15]. The parallel regulator operated at a frequency of 140 kHz and generated a 12 V output from an input range of 250–400 V. The converter also attained a 94 % efficiency [16]. The converter utilizing zero-voltage-transition technology achieved an efficiency of 95.5 % when supplied with an input voltage of 220 V and produced an output voltage of 400 V, resulting in a rated power output of 1000 W [17]. The converter, which relied on an auxiliary power unit, functioned at a frequency of 100 kHz and produced an output voltage of 350 V. The converter also attained a 90 % efficiency [18].

The researchers have utilized a microcontroller (TMS320F28335)-based dc-dc boost converter in order to attain a notable efficiency of 96 %. While the researchers prioritized efficiency computation, they did not place significant attention on transient analysis [19]. The researchers utilized the Digital Signal Processor (DSP)-based Perturb and Observe (P&O) and Fuzzy Logic (FL) algorithms to achieve a maximum power output of 16.9 W. The researchers prioritized the concept of power, but they neglected to concentrate on the computation of efficiency [20]. The researchers have implemented a microcontroller (TMS320F28027) using Particle Swarm Optimization (PSO) and a boost converter to ensure steady voltage, settling time, and overshoot values. The converter successfully attained a consistent voltage of 60 V, with a settling time of 110 ms. The researchers exclusively concentrated on transient analysis, neglecting to prioritize steady state analysis [21]. The researchers have utilized a microcontroller-based converter to achieve a 3 % overshoot value. The researchers placed particular emphasis on transient analysis, while neglecting to prioritize steady state analysis [22]. The researchers have developed a neuro-fuzzy inference system based on a microcontroller (TMS320F240) in order to attain optimal efficiency. While the researchers prioritized efficiency computation, they did not place attention on transient analysis [23]. The researchers utilized a DSP (TMS320F28335)-based PID controller with a boost converter to achieve a stable voltage of 20 V and rapid transient reactions. The researchers prioritized the examination of transitory reactions, while neglecting the analysis of steady state responses [24]. The researchers have utilized a digital signal processing (DSP)-based converter to achieve a maximum power output of 500 W. The researchers prioritized the concept of power, but they did not give significant attention to the computation of efficiency [25].

The researchers utilized a DSP-based buck-boost converter to achieve a steady voltage of 12 V and enhance stability responses. The researchers prioritized stability responses, but they did not place attention on transient responses [26]. The researchers have developed a converter based on digital signal processing (DSP) technology, which incorporates a fuzzy logic controller to achieve a constant voltage of 350 V. While the researchers placed emphasis on stability responses, they did not specifically address transient responses

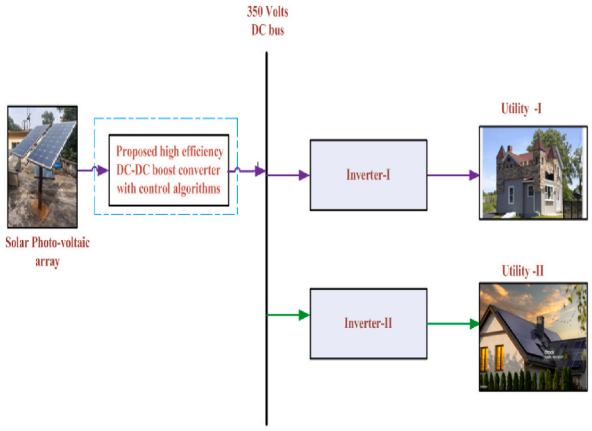


Fig. 2. Block diagram of the typical application of proposed converter.

[27]. The researchers have utilized a microcontroller-based converter with PI sliding mode control to attain a precise overshoot of 0 % and a rapid settling time of 30 ms. The researchers primarily investigated transient responses, but they did not place significant attention on steady state responses [28]. The researchers utilized a microcontroller-powered interleaved converter to attain rapid transient responses and a constant voltage of 350 V. The researchers specifically examined the transient responses, but they did not specifically examine the steady state responses [29]. The researchers have utilized DSP (F28379) powered power-consuming converters (PCC) and power-delivering converters (PDC) to achieve a consistent voltage of 78.5 V. The researchers primarily concentrated on stability analysis; however, they did not place much attention on transient analysis [30]. The researchers utilized three switch-based boost converters with FPGA and DSP controllers to achieve a power output of 30 W, as specified. The researchers prioritized power computation, but they did not place much emphasis on efficiency [31]. The converter achieved an output voltage of 350 V from an input voltage of 20 V, operating at a frequency of 1 kHz. The authors of this research have utilized FPGA-based PID and fuzzy logic controllers in the suggested converter to achieve a stable voltage of 350 V, a rise time of 10.624 ms, an undershoot value of 1.99 %, and a steady state error value of 0.0584.

The literature assessment suggests that researchers in the field of DC-DC converters have not prioritized or utilized transformer-less operation in inverters. In addition, the utilization of fuzzy logic controllers to enhance the performance characteristics of DC-DC converters was not applied. This particular difficulty that has been highlighted represents a gap in the existing research, which is the focus of this study. To address the research gap, one can put an inductor on the secondary side of the transformer used in the H-bridge of the DC-DC boost converter architecture. Additionally, suitable controllers can be incorporated in the feedback of the closed-loop control system. The suggested converter utilizes the Zero Current Switching (ZCS) technique to achieve gentle starting, minimize switching losses, and recover the energy stored in the magnetics during each half cycle. Additionally, the performance attributes of the suggested converter are examined and improved by using conventional (PI and PID) and Fuzzy logic controllers in a closed-loop system design. Fig. 2 illustrates the block diagram of the typical application of the suggested converter.

The contributions of the proposed converter are:

- ➢ Boost the DC link voltage to the level of Vmax in the output AC voltage obtained from inverter circuits.
- ➢ Elevate the DC voltage generated by renewable sources while effectively regulating it against line and load fluctuations for inverter application.
- ➢ Enable the inverters for transformer-less power conversion from DC to AC to reduce magnetic losses, size and weight of the inverter circuits used in the utility application.
- ➢ Th proposed converter's topology and switching sequences play a crucial role in enhancing overall performance.
- ➢ Zero Current Switching (ZCS) operation of the proposed converter recovers the stored energy from the magnetics.
- ➢ Hardware implementation of the proposed converter using FPGA based PID and Fuzzy logic controllers provide better voltage regulation, fast transient response and dynamic steady-state response.
- ➢ Simulation and experimental results are compared and analysed for the better matching of transient and steady state values.

This article has seven sections. Topology of the proposed converter is presented in section II. In section III, the design of PID and Fuzzy logic controllers are proposed. Implementation of the proposed converter is explained in section IV. Simulation and hardware results are discussed in sections V and VI. Finally, the article is concluded in section VII.

2. Topology of the proposed converter

Based on the above typical application schematic, the authors like to emphasis on proposed converter implemented with control algorithms. The proposed converter acts as a key element of the application. The topology of the proposed converter is depicted in

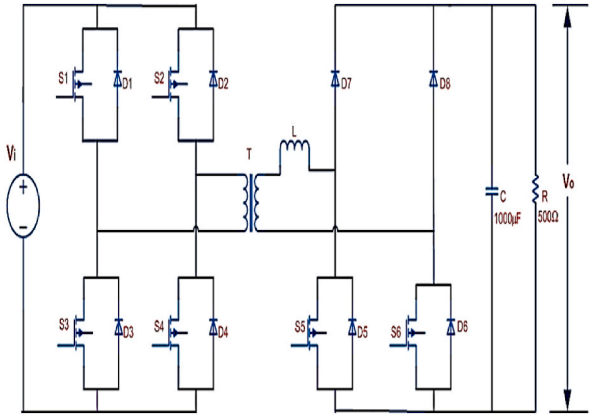


Fig. 3. Topology of the proposed converter [13].

Fig. 3, where V_i represents the input DC voltage source, S_1 to S_6 are active switches, D_1–D_6 correspond to the body diodes of switches S_1 to S_6, D_7 and D_8 serve as power diodes, C denotes the filter capacitor, T signifies the linear transformer, L stands for inductance, and R represents the load resistance. A power switch serves the dual purpose of providing current protection and establishing an electrical connection from a voltage source to a load. The diodes D_1–D_6, associated with the switches, create a path for load current, when the switch is in the OFF condition. To elevate the input DC voltage from 20 V to the output DC voltage of 350 V, a booster transformer is incorporated in this converter. An inductor is utilized on the output side of the booster transformer to smooth the current waveform by storing energy. Diodes D_7 and D_8 are employed to prevent current flow in the reverse direction. The filter capacitor C is implemented to reduce the ripples of the rectified voltage.

The equivalent circuits of the transformer used in the DC-DC boost converter for different periods are depicted in Fig. 4. A current path is created closely when S_2, S_3, and S_5 are turned ON during the period $[t_0\text{-}t_1]$. Similarly, the positions of the switches during $[t_1 - t_2]$, $[t_2 - t_3]$ and $[t_3 - t_4]$ are depicted in Table 1. Also, it can be observed that the transformer primary voltage ($\boldsymbol{V_p}$) is the input voltage (V_i) i.e., $\boldsymbol{V_P} = \boldsymbol{V_i}$, and the transformer secondary voltage ($\boldsymbol{V_{Sr}}$) is zero i.e., $\boldsymbol{V_{Sr}} = \mathbf{0}$ and therefore, the inductance voltage, $\boldsymbol{V_{L=}}(\boldsymbol{V_P} - \boldsymbol{V_{Sr}}) = (\boldsymbol{V_i} - \mathbf{0}) = \boldsymbol{V_i}$. With the help of $\mathbf{V_L} = \mathbf{L}\left(\frac{dI_L}{dt}\right)$.

and $\mathbf{V_L} = \mathbf{V_i}$ where L is the transformer parameter which represents the leakage inductance and thus

$$\mathbf{I_L} = \frac{\mathbf{V_i}}{\mathbf{L}}\mathbf{t} \tag{1}$$

During the time period $[t_1 - t_2]$ the switches in ON position are S_2 and S_3 while S_5 is OFF position. The diode D_7 conducts the current $\mathbf{I_L}$ which is the inductor current as a result of the OFF position of S_5. Further, it can be found as $\mathbf{V_P} = \mathbf{V_i}$, the secondary transformer voltage is equal to its output voltage (V_{Or}) i.e., $\boldsymbol{V_{Sr}} = \boldsymbol{V_{Or}}$ and then the inductance voltage is $\boldsymbol{V_L} = \boldsymbol{V_P} - \boldsymbol{V_{Sr}} = \boldsymbol{V_i} - \boldsymbol{V_{Or}}$.

Therefore, the current through the transformer at time **t1** (I_{T1}) can be derived by

$$\boldsymbol{I_L} = \frac{\boldsymbol{V_i} - \boldsymbol{V_{Or}}}{\boldsymbol{L}}\boldsymbol{t} + \mathrm{I_{T1}} \tag{2}$$

During the time period $[t_2 - t_3]$, switch S_2 is in OFF position and S3 is in ON position. The diode D_4 conducts $\boldsymbol{I_L}$ due to OFF position of S_2. Therefore, the inductance current using inductor current at time $\boldsymbol{t2}$ ($\boldsymbol{I_{T2}}$) is derived by

$$\mathbf{I_L} = \mathbf{I_{T2}} - \frac{\mathbf{V_{Or}}}{\mathbf{L}}\mathbf{t} \tag{3}$$

During the duration $[\boldsymbol{t_3} - \boldsymbol{t_4}]$, there is no current flow in the converter since S_3 is in OFF position at $t = t_4$. The proposed converter is operated at a fixed switching frequency. Hence during the time periods T_1 and T_2, the current passes through the transformer is controlled which directly influences the power of the converter. Also, S_5 and S_6 have been turned ON for T_1, with difference of 180° in phase. On the other hand, S_1 and S_2 are turned ON for a period equal to the sum of T_1 and T_2 with a difference of 180° in phase. The S_3 and S_4 are in ON for half switching time period, T/2, with the same phase difference of 180°.The equivalent circuits of the new topological converter during different time periods (t_1-t_8) are plotted in Fig. 4 for the first half cycle from (a) t_0-t_1 (b) t_1-t_2 (c) t_2-t_3 (d) t_3-t_4. The transformer inductance current $I_{T1} + I_{T2}$ is calculated based on the above equations. equations (1)–(3) represents the transformer inductance current. Since the LHS of equations (1)–(3) are equal, the RHS of the above equations can be equated. After equating, transformer time 1 current and time 2 current can be determined as,

$$\frac{\boldsymbol{V_i t}}{\boldsymbol{L}} = \frac{\boldsymbol{V_i} - \boldsymbol{V_{Or}}}{\boldsymbol{L}}\boldsymbol{t} + \boldsymbol{I_{T1=}I_{T2}} - \frac{\boldsymbol{V_{Or}}}{\boldsymbol{L}}\boldsymbol{t} \tag{4}$$

The transformer is operated with a switching frequency of 1 kHz, i.e., a time period of 1 ms. Based on the parameters of input

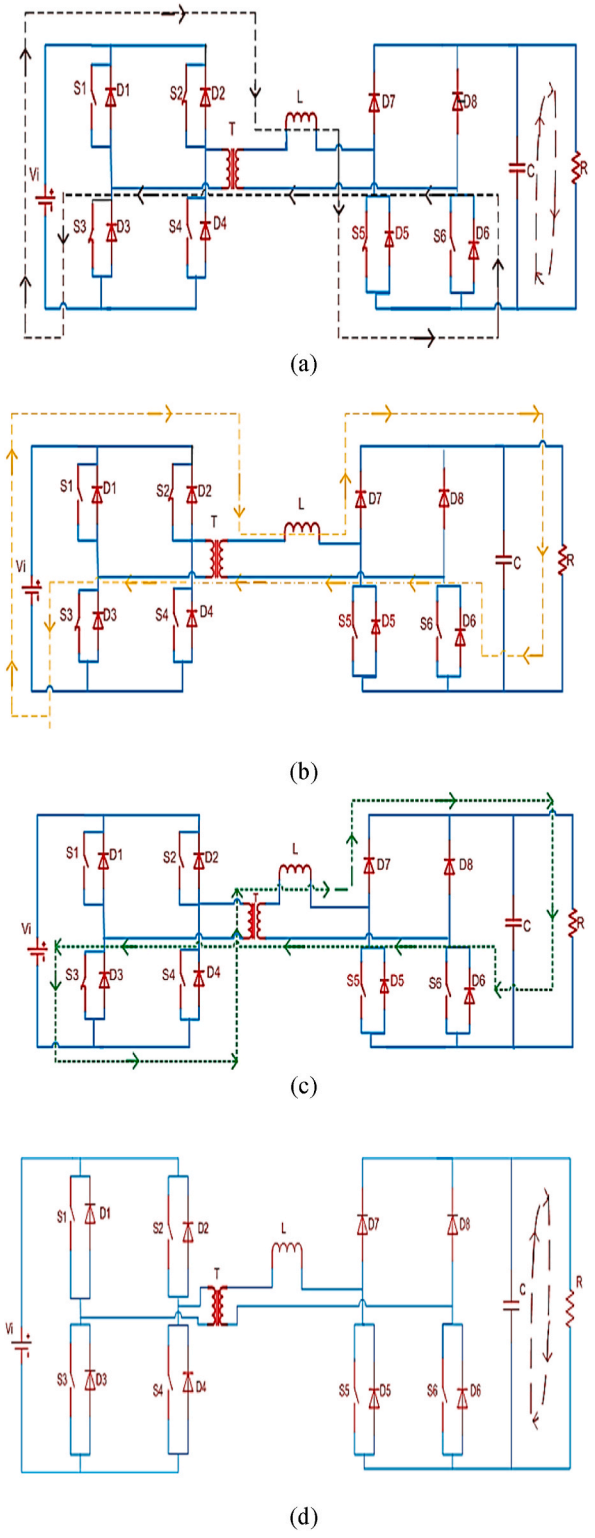


Fig. 4. Equivalent circuits of the converter's operation during different time periods in the first half cycle (a) t_0-t_1 (b) t_1-t_2 (c) t_2-t_3 (d) t_3-t_4.

Table 1
Time duration and status of switches.

Time duration	Switches in ON	Switches in OFF
t_0-t_1	S_2, S_3, S_5	S_1, S_4, S_6
t_1-t_2	S_2, S_3	S_1, S_4, S_5, S_6
t_2-t_3	S_3	S_1, S_2, S_4, S_5, S_6
t_3-t_4	Nil	S_1, S_2, S_3, S_4, S_5, S_6

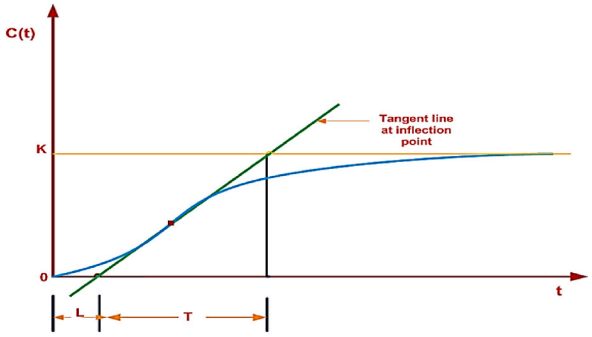


Fig. 5. Response of converter for unit step input [32].

Table 2
ZIEGLER–NICHOLS tuning rule based on step response.

Controller	K_p	T_i	T_d
P	T/L	∞	0
PI	0.9 T/L	$L/0.3$	0
PID	1.2 T/L	2L	0.5L

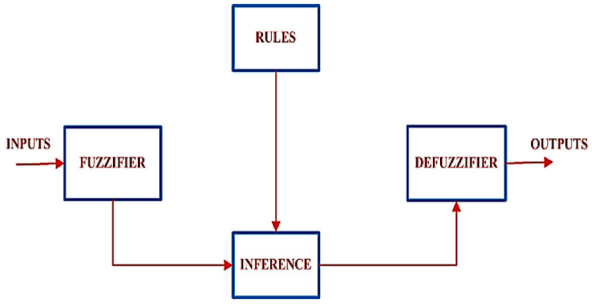
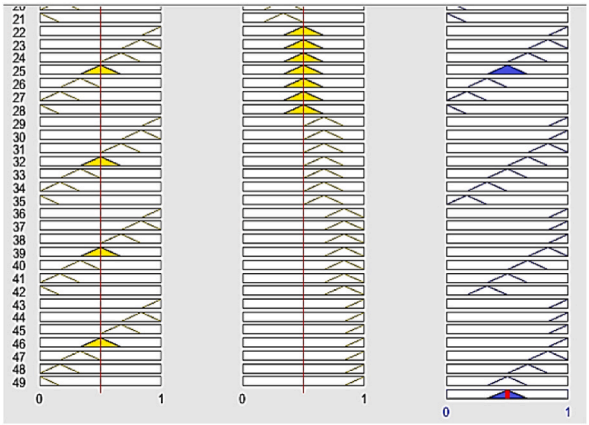


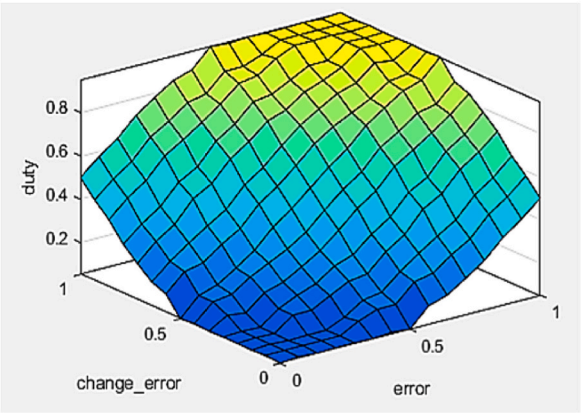
Fig. 6. Architecture of Fuzzy logic controller.

Table 3
Rule based fuzzy decision map

Error	Change in Error						
	NB	NM	NS	Z	PS	PM	PB
PB	Z	NS	NM	NB	NB	NB	NB
PM	PS	Z	NS	NM	NB	NB	NB
PS	PM	PS	Z	NS	NM	NB	NB
Z	PB	PM	PS	Z	NS	NM	NB
NS	PB	PB	PM	PS	Z	NS	NM
NM	PB	PB	PB	PM	PS	Z	NS
NB	PB	PB	PB	PB	PM	PS	Z



(a) Rules of the fuzzy system for the input and output variables



(b) Surface of the fuzzy system for the input and output variables

Fig. 7. Output representation of the Fuzzy system: (a) Rules of the Fuzzy system for the input and output variables (b) Surface of the Fuzzy system for the input and output variables.

Table 4
Classification of membership functions with their abbreviations

S. No	Membership functions	Abbreviations of membership function
1	trimf	Triangular membership function
2	trapmf	Trapezoidal membership function
3	gbellmf	Generalized bell shaped membership function
4	gaussmf	Gaussian membership function
5	gauss2mf	Gaussian 2 membership function
6	sigmf	Sigmoidal membership function
7	dsigmf	Difference between two sigmoidal membership function
8	psigmf	Product of two sigmoidal membership functions
9	pimf	Pi-shaped membership function
10	smf	Spline-based S-shaped membership function
11	zmf	Spline-based Z-shaped membership function

Table 5
Specifications of the proposed converter.

Parameter	Values for converter
Input Voltage	20 V
Linear Transformer	Nominal Power = 350 W Frequency = 1000 Hz $V_{1rms=}$20 V, V_{2rms} = 355 V
Capacitor (C)	1000 μF
Inductor (L)	1 mH
Load Resistance (R)	500 Ω
Switching frequency	1 KHz
K_p	10
K_i	0.1
K_d	0.01

voltage, transformer output voltage, time duration of each phase, inductor value, and the transformer currents I_{T1} and I_{T2}, the transformer inductance current can be calculated.

$$V_i t - V_i t + V_{or} t - I_{T1} L - I_{T2} L + V_{or} t = 0 \tag{5}$$

$$I_{T1} + I_{T2} = \frac{2V_{or} t}{L} \tag{6}$$

3. Design of PID and fuzzy logic controllers

The controllers are used to compensate for the variations in the converter [32]. The traditional controllers PI and PID are linear controllers. Similarly, the Fuzzy logic controller acts as a non-linear controller. Since the switching converters exhibit non-linear phenomena, both linear and non-linear controllers are used for determining the performance metrics of the converter [5]. The controllers are necessary for the converter to compensate the line and load variations and provide the better voltage regulation.

3.1. PI and PID controllers

In PI control, large disturbances and noises are avoided. This controller is more complicated to tune [32]. This is the most popular feedback control without offset values. The output (U) of the PI controller is given by:

$$\mathbf{U} = \mathbf{k_p} + \mathbf{k_i} \int_0^t \mathrm{edt} \tag{7}$$

Where k_p is the proportional gain, k_i is the integral gain, and e is the error. In PID Control, settling time is decreased. This controller is the most complicated to tune. Derivative action may be affected by noise without offset values. The response of the PID controller is given by:

$$\mathbf{U} = \mathbf{k_p} + \mathbf{k_i} \int_0^t \boldsymbol{edt} + \mathbf{k_d} \frac{\mathrm{de}}{\mathrm{dt}} \tag{8}$$

where k_d is the derivative gain. Also, it is used to control magnitude and phase problems. The PID tuning can be done by two methods, namely (i) the transfer function method and (ii) the Ziegler and Nichols method (practical method). The proposed converter consists of many switching and operating states, so the implementation of PID controller using transfer function seems to be challenging. The Ziegler and Nichols method seems to be convenient and widely used because it is a practical method. The Ziegler and Nichols method (first method) is used for the tuning of the proposed converter by the two constant values of delay time L and time constant T [32].

Ziegler and Nichols proposed two rules that are used for determining the values of the proportional gain, integral time, and derivative time based on the transient response characteristics of a converter. In the first rule, the response of the converter is determined experimentally by applying it to a unit-step input, as depicted in Fig. 5. If a step input to a converter produces an S-shaped response curve, then the first rule is applicable. The output of the converter may be done experimentally or from a dynamic simulation. The S-shaped response curve has two constants: delay time L and time constant T. The two constants are obtained by drawing a tangent line at the inflection point of the S-shaped response curve and determining the intersections of the tangent line with the time axis and line $c(t) = K$, *as* depicted in Fig. 5. The Ziegler and Nichols tuning rules for linear controllers are determined using Table 2 [32].

Based on Table 2, the PID values can be calculated using the two constants T and L. $K_p = 1.2\frac{T}{L}$, $T_i = 2L$ and $T_d = 0.5L$. By substituting the values of L = 1 and T = 2, the K_p, T_i and T_d values are calculated as 2.4, 2sec and 0.5 s. The tuning of PID controller by the first method of Ziegler –Nichol's rule is expressed as,

$$\boldsymbol{G_C}(\boldsymbol{s}) = \boldsymbol{K_P}\left(1 + \frac{\mathbf{1}}{\boldsymbol{T_i s}} + \boldsymbol{T_d s}\right) \tag{9}$$

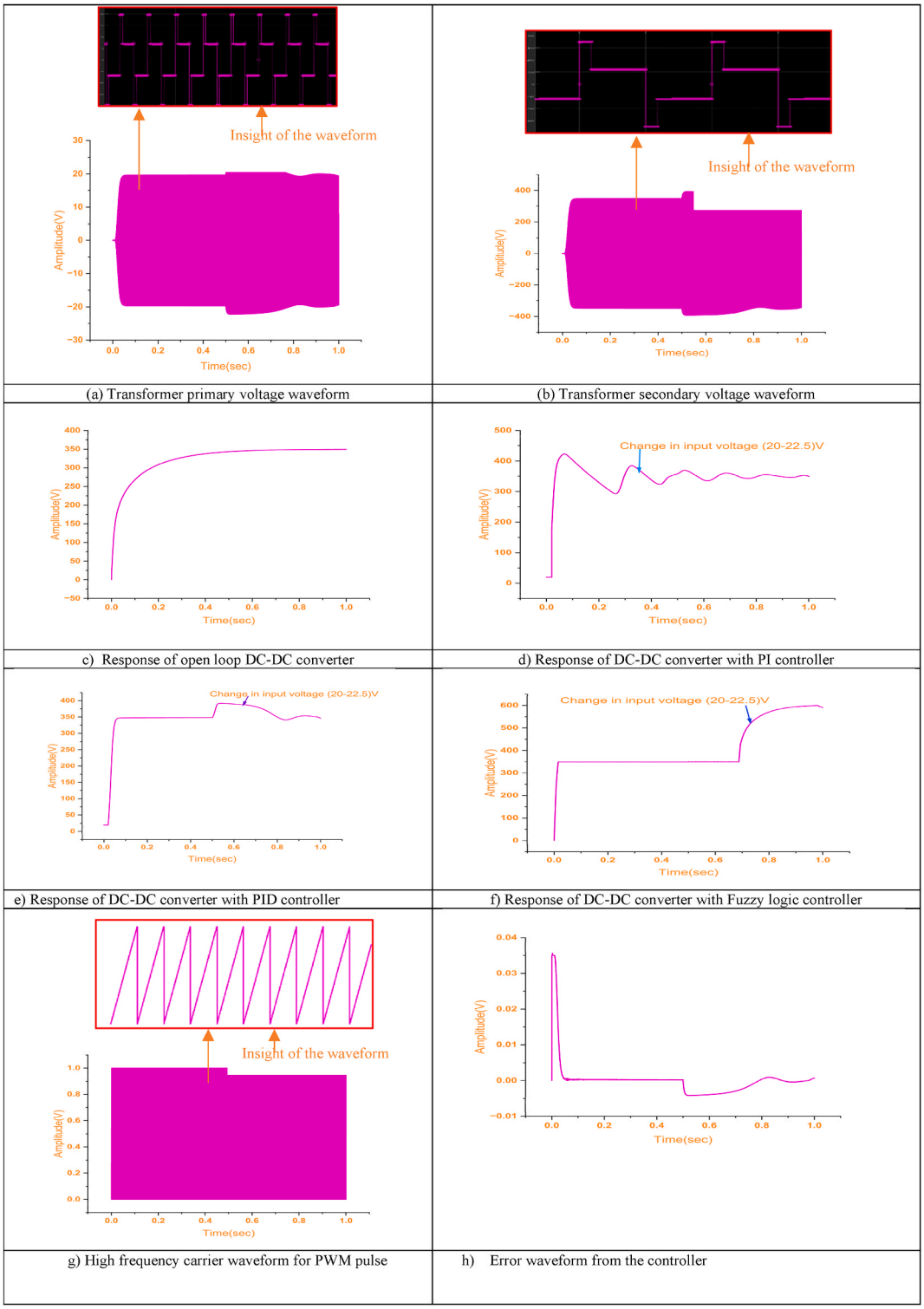


Fig. 8. Waveforms of (a) transformer primary and (b)secondary voltage and (c), d), (e), (f)are the responses of the proposed converter with and without controllers, (g) High-frequency carrier waveform for PWM pulse and (h)Error waveform measured from the controller.

$$G_C(s) = 1.2\frac{T}{L}\left(1 + \frac{1}{2Ls} + 0.5Ls\right) \tag{10}$$

Table 6
Comparison OF PI controller based step time characteristics between 3 converters.

Characteristics	PI Controller		
	Proposed converter	Adaptive control method [34]	Current-mode controller [35]
Rise Time (ms)	4.514	59	15
Settling Time (ms)	9.105	7.9	39.5
Undershoot (%)	1.992	3.61	2

Table 7
Comparison of simulation-based step time characteristics with and without controller for different load resistances and input voltages of proposed converter.

Controller	Characteristics	$R_L = 350\Omega$ $V_{in} = 15$ V	$R_L = 350\ \Omega$ $V_{in} = 20$ V	$R_L = 350\ \Omega$ $V_{in} = 25$ V	$R_L = 250\ \Omega$ $V_{in} = 20$ V	$R_L = 500\Omega$ $V_{in} = 20$ V
Open Loop	Steady-state error	NA	NA	NA	NA	NA
	Rise Time (ms)	217.030	224.448	211.943	148.064	382.005
	Undershoot (%)	2.223	2.251	2.215	2.314	2.178
	Overshoot (%)	0.872	0.981	0.810	0.886	0.797
	Slew rate (v/ms)	540.419	623.881	992.732	917.349	434.867
PID	Steady-state error	0.0225	0.00039	0.00078	0.00036	0.00043
	Rise Time (ms)	63.535	93.449	56.179	112.649	88.781
	Undershoot (%)	2.000	2.251	2.000	2.000	2.000
	Overshoot (%)	−0.648	9.341	8.152	8.152	9.341
	Slew rate (v/ms)	1.691	2.801	4.664	2.330	2.970
Fuzzy	Steady-state error	0.1983	0.5	0.5	0.5	0.0584
	Rise Time (ms)	19.982	9.384	9.385	9.427	10.624
	Undershoot (%)	1.984	1.880	1.880	1.880	1.990
	Overshoot (%)	0.503	0.505	0.505	0.504	0.505
	Slew rate (v/ms)	13.801	29.502	32.768	26.998	26.037

$$G_C(s) = 0.6\mathrm{T}\ \frac{\left(s+\frac{1}{L}\right)2}{s} \tag{11}$$

3.2. Fuzzy logic controller

In addressing non-linearity control issues, controllers based on Fuzzy logic can be employed to regulate the response of the converters [33]. The robust nature of the Fuzzy logic controller makes it well-suited for handling non-linear problems. Fuzzy logic also provides fast dynamic response. The Fuzzy logic rules can be implemented without the knowledge of system parameters. Fuzzy logic operation necessitates the use of the error and change in error of the converter's response as inputs. Generally, the disparity between the actual output and the predicted one is referred to as an error. In the context of this work, focusing on voltage control, the error can be calculated as the difference between the actual voltage and the reference voltage. Additionally, the change in error, representing the difference in error from the present state to the past state, is defined as follows:

$$\mathbf{e}(\mathbf{V}) = \mathbf{V_{out}}\text{-}\mathbf{V_{ref}} \tag{12}$$

$$\mathbf{d}\{\mathbf{e}(\mathbf{V})\} = \mathbf{e}(\mathbf{V})_2\text{-}\mathbf{e}(\mathbf{V})_1 \tag{13}$$

where *e(V)* is the error voltage of the converter, $\mathbf{V}_{\text{out}}$ is the actual output voltage, $\boldsymbol{V_{ref}}$ is the reference voltage, herein it is 350 V, $\mathbf{d}\{\mathbf{e}(\mathbf{V})\}$ is the change in error, $\mathbf{e}(\mathbf{V})_1$ is the voltage error at state 1, and $\mathbf{e}(\mathbf{V})_2$ is the error voltage at state 2. Also, the main advantage of the controller based on fuzzy logic is that it needs only the linguistic type variables as input and not the numerical type variables. The elements of a Fuzzy logic controller are a fuzzifier, a rule base and a de-fuzzifier. In fuzzification, the fuzzy set is created using the crisp set. The block diagram of the Fuzzy logic controller is given in Fig. 6.

The types of the MAMDANI interference with seven linguistic variables is employed for performing the voltage control. The seven variables are Negative Big (NB), Negative Medium (NM), Negative Small (NS), Zero (Z), Positive Small (PS), Positive Medium (PM) and Positive Big (PB). Hence, there has been a rule base formed with forty-nine rules for the seven linguistic variables. The error and the change in error for the respective linguistic variables are depicted in Table 3.

The Fuzzy logic controller toolbox present in Simulink consists of a Fuzzy logic designer window. The Fuzzy inference system (FIS) present in the Fuzzy logic designer is of two types, like Mamdani and sugeno. The variables for FIS are input and output variables. The input variables are error and change in error (CE). Then the output variable is the duty cycle. The types of membership functions are trimf, trapmf, gbellmf, gaussmf, gauss2mf, sigmf, dsigmf, psigmf, pimf, smf, and zmf. The rules and surface of the Fuzzy system based

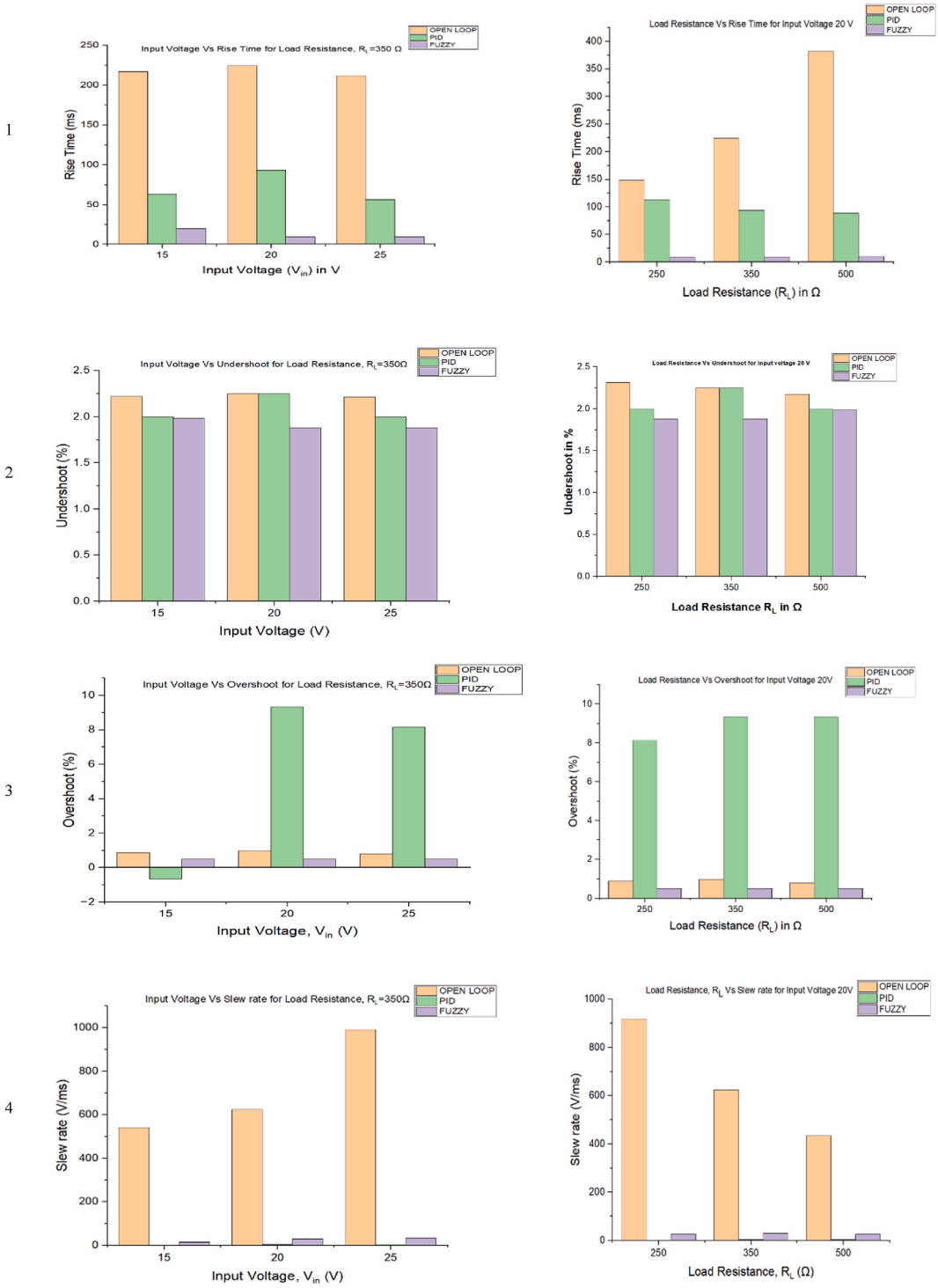


Fig. 9. Comparison of the simulation-based performance metrics of the proposed converter in terms of 1) rise time, 2) undershoot, 3) overshoot, and 4) slew rate. The left column is variations of input voltage, and the right column is variations of load resistance for open-loop, PID, and fuzzy logic controllers.

Table 8
Simulation results of line analysis of proposed converter for open loop, PID, and fuzzy controllers for load resistance $R_L = 500\ \Omega$

	Input voltage(V)	Input current(A)	Output current(A)	Output voltage(V)	Input power(W)	Output power(W)
Open loop	23	21.68	0.9155	457.6	498.64	418.93
	22	20.73	0.8754	437.6	456.06	383.07
	20	18.84	0.7953	397.6	376.8	316.21
	18	16.94	0.7152	357.5	304.92	255.68
	16	15.05	0.6352	317.5	240.8	201.67
PID	23	15	1.05	349.9	345	367.39
	22	10.39	1.049	350	228.58	367.15
	20	15.81	1.05	349.9	316.2	367.39
	18	20.33	1.05	350	365.94	367.5
	16	0.0007	0.38	129.5	0.0112	49.21
Fuzzy	23	18.5	0.987	350	425.5	345.45
	22	18	0.896	350	396	313.6
	20	17.3	0.74	350	346	259
	18	17.2	0.72	350	309.6	252
	16	17	0.7	320	272	224

Table 9
Simulation results of load analysis of proposed converter for open loop, PID, and fuzzy controllers for input voltage $V_I = 20$ V.

	Load resistance(Ω)	Input current(A)	Output current(A)	Output voltage(V)	Input power(W)	Output power(W)
Open loop	350	20.11	1.001	350.1	402.2	350.45
	370	19.93	0.9652	357	398.6	344.57
	400	19.66	0.9178	367	393.2	336.83
	430	19.4	0.876	376.6	388	329.90
	470	19.07	0.8274	368.8	381.4	305.14
	500	18.84	0.7953	397.6	376.8	316.21
PID	460	13.12	1.111	350	262.4	388.85
	470	15.78	1.095	350	315.6	383.25
	480	10.69	1.079	350	213.8	377.65
	490	11.22	1.064	350	224.4	372.4
	500	15.81	1.05	349.9	316.2	367.39
Fuzzy	480	17.89	0.98	350	358	343
	490	17.76	0.87	350	355.2	304.5
	500	17.65	0.86	350	353	301
	510	17.54	0.88	350	350.8	308
	520	17.38	0.85	350	347.6	298

Table 10
Hardware components of the proposed converter.

Components	Specifications
Switches	SIC MOSFET C3M0032120D-1200V, 63A,
Linear Transformer	355V, 350W
Inductor	100 μH,30A
Diode	Hyperfast Diode RHRG30120-12000V,30A
Electrolytic Capacitor	680 μF,450V
Variable Rheostat	500 Ω,2A

on input and output variables are depicted in Fig. 7. The classification of membership functions is depicted in Table 4.

4. Implementation of the proposed converter with controllers

The design and analysis of the proposed converter have been performed by simulation using SIMULINK/MATLAB 2021R installed in a system configured with an Intel i3 processor of the 11th generation and 8 GB of RAM. The input parameter values for the proposed converter are given in Table 5.

The voltage of the response of the converter is compared with the reference voltage (350 V) to estimate the error voltage. Thereafter, the error is corrected using the PID and Fuzzy logic controller separately as stated above, which are connected as feedback to the converter in a closed loop. The final output of the converter with each controller in feedback is compared using two comparators with a saw-tooth signal, whose frequency is high and equal to the frequency of the reference voltage for generating two pulse width modulated (PWM) signals of pulse widths T_1 and T_2 which can be used as control signals. Also, the difference between a constant

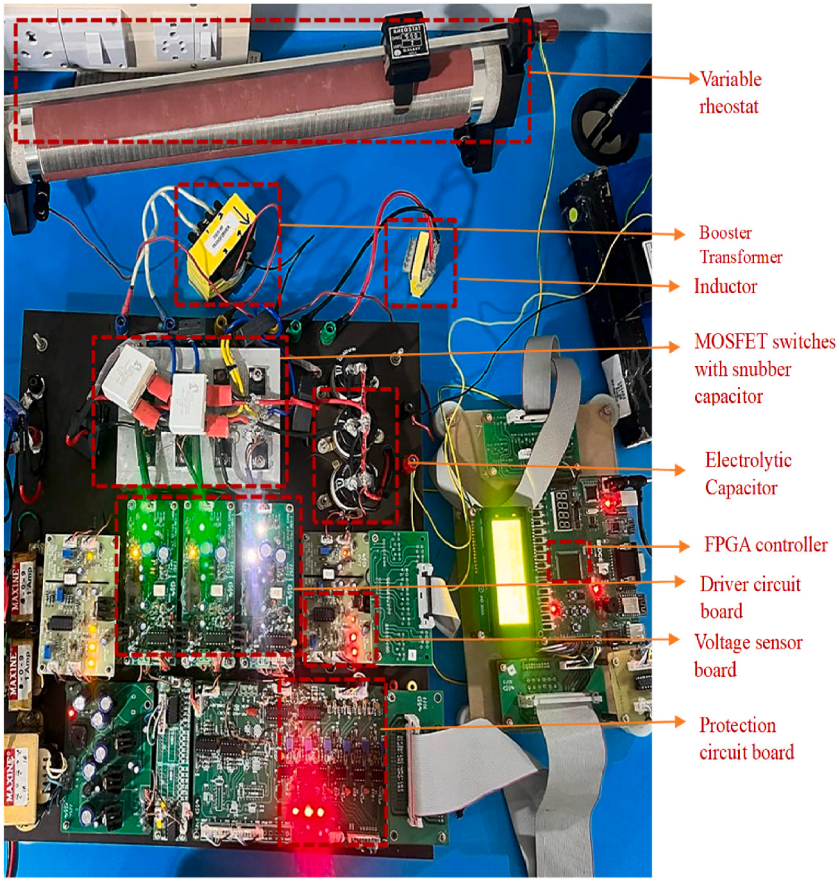


Fig. 10. Hardware realization of the proposed converter.

voltage signal and the saw tooth signal is estimated to generate one more PWM signal whose pulse width is $T_{1/2}$.

5. Simulation results and discussion

The proposed converter is simulated with an input of 20 V and produces an output of 350 V. The remaining input parameters for the proposed converter are given in Table 5. The constant values for proportional, integral, and derivative controllers, respectively, are set at 10, 0.1, and 0.01 based on the trial-and-error method. The resultant waveforms obtained from simulation are shown in Fig. 8. The waveforms for transformer primary and secondary voltages are depicted in Fig. 8(a) and (b), respectively. Fig. 8(c) is the waveform of the converter without a controller. Fig. 8(d), (e), and (f) are the voltage waveforms of the converter with PI, PID, and Fuzzy logic controllers, respectively. The high-frequency carrier waveform and error waveform from the controller are depicted in Fig. 8(g) and (h). The step input is given as a voltage disturbance at times t = 0.5 and t = 0.9 s to analyse the performance of the converter with the controller.

Table 6 describes the step response characteristics of rise time, settling time, and undershoot of a fixed current-mode controller, an adaptive controller, and proposed converter method. The fixed current-mode controller method has a rise time of 15 ms, a settling time of 39.5 ms, and an undershoot of 2 %. The adaptive control method has a rise time of 59 ms, a settling time of 7.9 ms, and an undershoot of 3.61 %. The proposed converter method has a rise time of 4.514 ms, a settling time of 9.105 ms, and an undershoot of 1.992 %. The comparison between these three methods determines that the proposed converter has less rise time and lower undershoot values. But the settling time of the proposed converter is moderate compared to the other two methods. The measured values of the characteristics of the converter with controllers for the given step function are tabulated in Table 7. It can be observed that the Fuzzy logic controller outperforms in terms of rise time overshoot, undershoot, and slew rate compared to the PID controllers.

The time-domain specifications such as rise time, overshoot, undershoot, and slew rate have been measured as in Table 7 and plotted in Fig. 9. The PID and Fuzzy logic controllers are used for voltage stability by varying resistance loads at 250 Ω, 350 Ω and 500 Ω. Similarly, the steady state error and time domain specifications have been measured for different loads and input voltages, as shown in Table 7 and depicted in Fig. 9. The simulation results of both load and line analysis are determined in Tables 8 and 9. The load analysis is performed by varying the different load resistances for both the with and without controller conditions. The load analysis is performed by maintaining the input voltage V_{in} = 20 V as a constant. The line analysis is performed by varying the different input

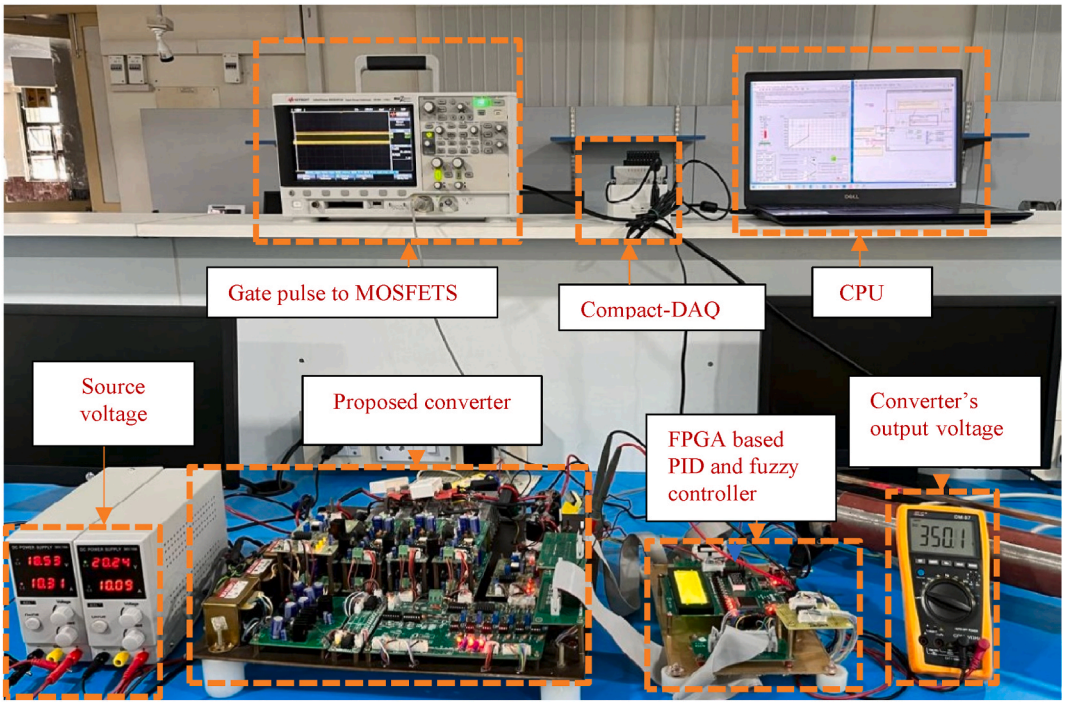


Fig. 11. Experimental setup of the proposed converter.

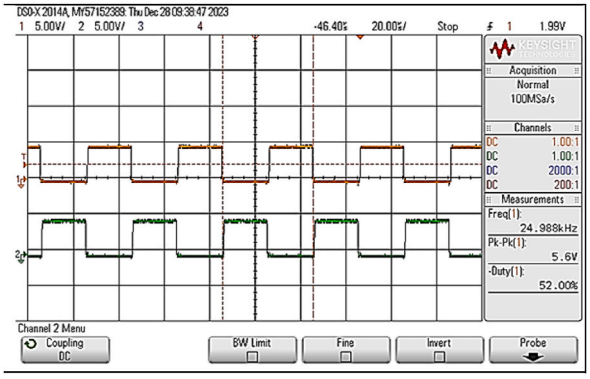


Fig. 12. Switching pulses to the proposed converter.

values and maintaining the load resistance $R_L = 500\ \Omega$ as a constant. The line analysis is also done for both the with and without controller conditions. By comparing the open-loop, PID, and Fuzzy logic controllers, the output voltage varies in the open-loop condition. But in PID and fuzzy logic controllers, the output voltage $V_o = 350V$ is maintained, even though the line values and load values are varied. The input and output powers are determined based on the input and output values. The load resistances R = 350Ω, 370Ω, 400Ω, 430Ω, 470Ω, and 500Ω are varied for the open loop system, and the load resistances R = 460Ω, 470Ω, 480Ω, 490Ω, and 500Ω are varied for the PID controller system. The load resistances R = 480Ω, 490Ω, 500Ω, 510Ω, and 520Ω are varied for the Fuzzy logic controller system. The output voltage and steady-state error are determined for different controllers: PID, fuzzy, and open-loop systems. Even though the load and input voltage vary, the output voltage almost remains constant. The controller in the feedback compensates for the variations in the load and produces a constant output voltage.

6. Hardware development

An EDGE Spartan 6 FPGA chip acts as the brain of the converter. This programmable controller uses algorithms to adjust for variations in input and output voltages. For increased efficiency and power handling, the prototype utilizes silicon-carbide MOSFET switches. A special type of transformer (linear transformer) boosts the voltage to a desired level of 355V, delivering up to 350W of

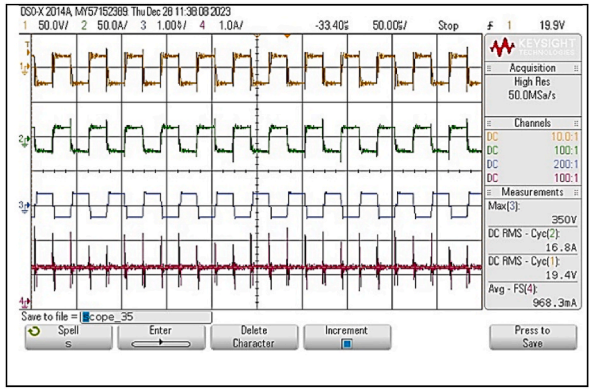


Fig. 13. Waveforms of voltage and current in the proposed converter's transformer.

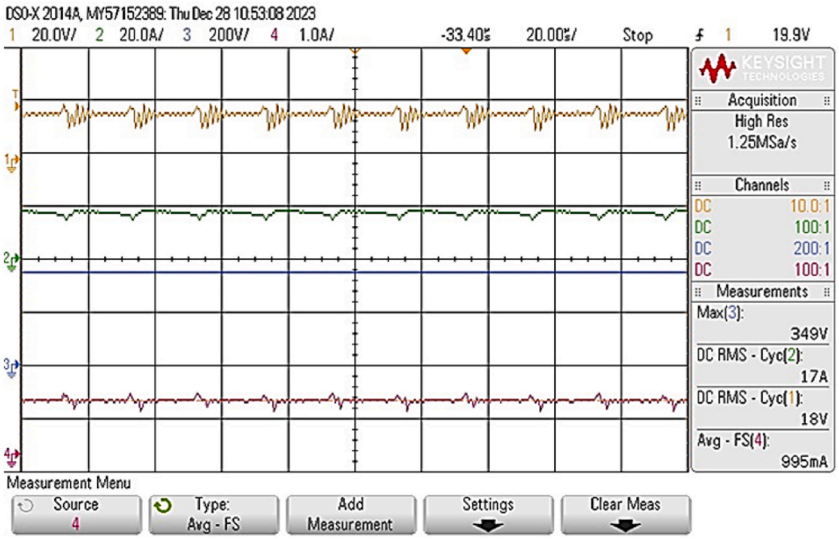


Fig. 14. Voltage and current waveforms of a proposed converter.

Table 11
Comparison of hardware-based step time characteristics with and without controller for different load resistances and input voltages of proposed converter.

Controller	Characteristics	$R_L = 350\Omega$ $V_{in} = 15$ V	$R_L = 350\ \Omega$ $V_{in} = 20$ V	$R_L = 350\ \Omega$ $V_{in} = 25$ V	$R_L = 250\ \Omega$ $V_{in} = 20$ V	$R_L = 500\ \Omega$ $V_{in} = 20$ V
Open loop	Rise Time (ms)	220.03	230.44	215.94	150.06	385.00
	Undershoot (%)	2.250	2.300	2.250	2.650	2.350
	Overshoot (%)	0.972	1.000	0.910	0.996	0.850
	Slew rate (v/ms)	600.41	650.881	998.73	950.34	469.86
PID	Rise Time (ms)	67.54	95.45	60.2	120.72	99.81
	Undershoot (%)	2.202	2.500	2.202	2.202	2.202
	Overshoot (%)	−0.86	10.35	9.22	9.22	10.5
	Slew rate (v/ms)	2.691	2.801	4.664	2.330	2.970
Fuzzy	Rise Time (ms)	22.68	12.58	12.69	12.70	15.95
	Undershoot (%)	2.000	2.000	2.020	2.035	2.000
	Overshoot (%)	1.002	1.303	1.40	1.623	1.000
	Slew rate (v/ms)	15.967	30.67	34.89	30.86	32.50

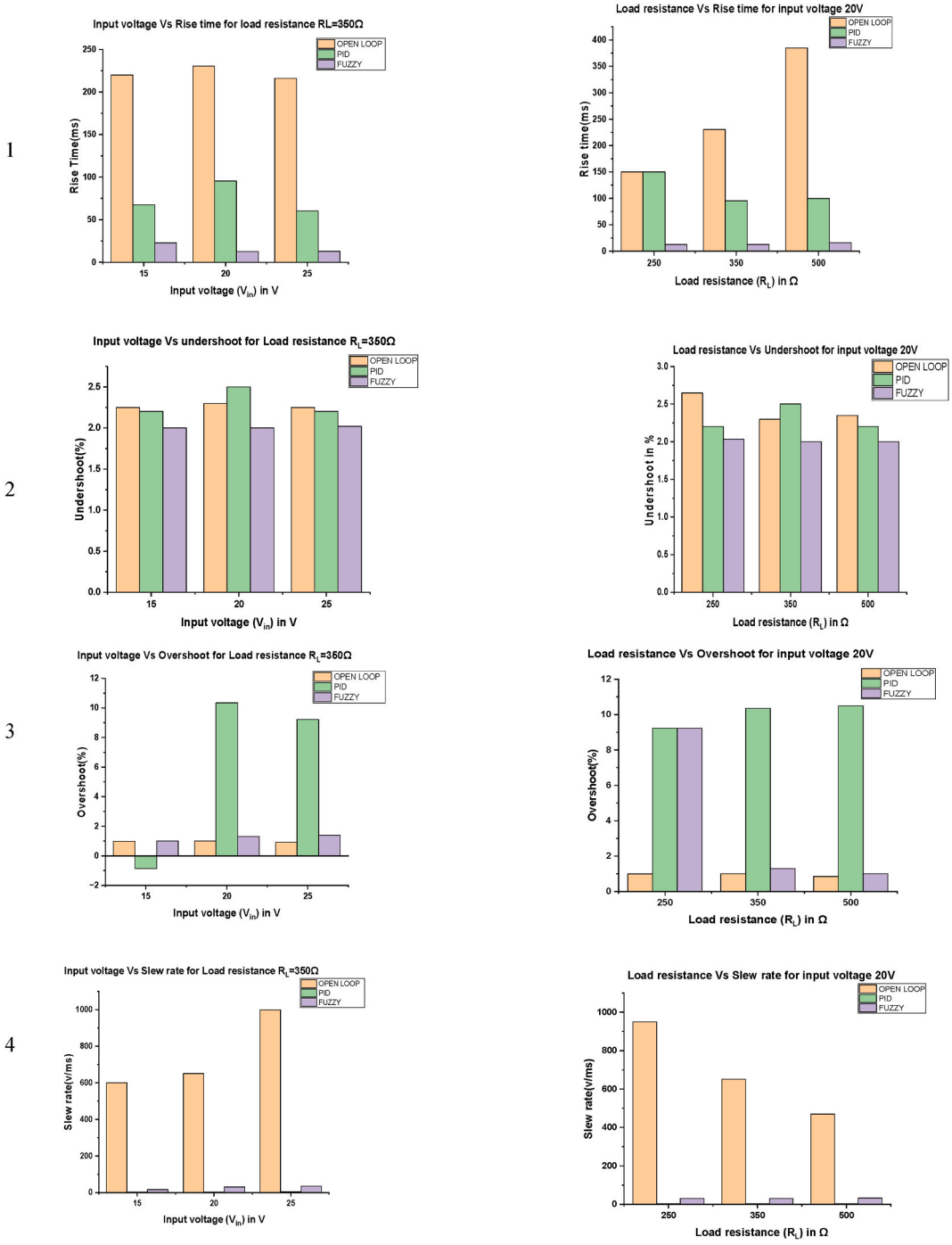


Fig. 15. Comparison of the hardware-based performance metrics of the proposed converter in terms of 1) rise time, 2) undershoot, 3) overshoot, and 4) slew rate. The left column is variations of input voltage, and the right column is variations of load resistance for open-loop, PID, and Fuzzy logic controllers.

power. An inductor with a value of 100 μH and a current rating of 30A smooths out the current flow. Finally, a capacitor rated at 680 μF and 450V filters any remaining voltage fluctuations. The load is simulated by a variable resistor (rheostat) with a resistance of 500Ω and a current capacity of 2A. The hardware components of the proposed converter are depicted in Table 10.

Fig. 10 shows the prototype converter with its components. An IC called a TL082 is used alongside a protective circuit board containing ICs 4027 and 4098. This combination senses various parameters like input and output current (I_{dc}, I_{out}), input and output voltage (V_{dc}, V_{out}), and load current (I_L). The FPGA controller then uses this information. The voltage sensing board acts as the

Table 12
Experimental results of load analysis for open loop, PID, and fuzzy for input voltage $V_I = 20$ V.

	Load resistance(Ω)	Input current(A)	Output current(A)	Output voltage(V)	Input power(W)	Output power(W)	η (%)
Open loop	350	18	1	321.6	360	321.6	89.33
	370	18	1	339.4	360	339.4	94.27
	400	17	1	330	340	330	97.05
	430	17	1	329	340	329	96.76
	470	17	1	334	340	334	98.23
	500	12	0.6	314	240	188.4	78.5
PID	460	19	0.7	350	380	245	64.47
	470	15	0.7	350	300	245	81.66
	480	17	0.7	350	340	245	72.05
	490	17	0.7	350	340	245	72.05
	500	15	0.7	350	300	245	81.66
Fuzzy	480	15	0.7	350	300	245	81.66
	490	14.9	0.7	350	298	245	82.21
	500	14.8	0.7	350	296	245	82.77
	510	15	0.7	350	300	245	81.66
	520	14.79	0.7	350	296	245	82.77

Table 13
Experimental results of line analysis for open loop, PID, and fuzzy for load resistance $R_L = 500$ Ω

	Input voltage(V)	Input current(A)	Output current(A)	Output voltage(V)	Input power(W)	Output power(W)	η (%)
Open loop	23	13	0.6	346.5	299	207.9	69.53
	22	12	0.6	327.8	264	196.68	74.5
	20	12	0.6	314	240	188.4	76.43
	18	10	0.5	276.8	180	138.4	76.88
	16	10	0.5	254.6	160	127.3	79.56
PID	23	12	0.7	350	276	245	88.76
	22	13	0.7	350	286	245	85.66
	20	14	0.7	350	280	245	87.5
	18	17	0.7	350	306	245	80.06
	16	17	0.6	320	272	192	70.58
Fuzzy	23	11.5	0.7	350	264.5	245	92.62
	22	13	0.7	350	286	245	85.66
	20	13	0.7	350	286	245	85.66
	18	15	0.7	350	270	245	90.74
	16	15	0.7	310	240	217	90.41

controller's feedback loop, closing the loop in the control code. Additionally, a Hall-effect sensor measures both AC and DC current. To protect the system from voltage spikes caused by sudden current changes, a snubber circuit is employed. This circuit limits unwanted power dissipation and counteracts the parasitic inductance inherent in electrical wiring.

The overall experimental setup of the proposed converter is depicted in Fig. 11. The source voltage provides the input voltage of 20 V to the proposed converter. The proposed converter boosts the voltage up to 350 V. The FPGA-based PID and Fuzzy logic controllers are implemented to maintain better voltage regulation and improve the performance metrics of the converter. The C-DAQ (Compact Data Acquisition) is interfaced with LabVIEW software for displaying the output values. The C-DAQ NI-9201 is used to acquire the converter output and display the transient output on the CPU.

Fig. 12 shows the pulses used to switch the converter on and off. Fig. 13 then depicts the measured voltages and currents at the transformer within the converter. The top waveform in Fig. 13 corresponds to the voltage on the primary side of the transformer, followed by the current on the primary side. The bottom waveform shows the voltage on the secondary side, and the final waveform shows the current on the secondary side.

The measured proposed converter's input and output voltages and current waveforms are depicted in Fig. 14. The top waveform in Fig. 14 corresponds to the input voltage 18 V of the proposed converter, followed by the input current 17 A. The bottom waveform shows the output voltage 349 V of the proposed converter, and the final waveform shows the output current 995 mA.

The experimental step-time characteristics of the proposed converter are depicted in Table 11 and plotted in Fig. 15. The characteristics are measured and analysed for both controllers and without controllers. Based on the comparison of open loop, PID, and Fuzzy logic controllers, the Fuzzy logic controller has better step-time characteristics than the open loop and PID controllers.

The experimental-based load and line analysis are performed and depicted in Tables 12 and 13. Similarly, the simulation-based load and line analysis are performed and depicted in Tables 8 and 9 Based on the comparison of simulated and experimental results, the graphs are depicted in Figs. 16 and 17 for input and output powers by varying different voltages and load resistances. In Fig. 16, the load analysis of both simulation and experimental values of proposed converter-based open-loop, PID, and fuzzy logic

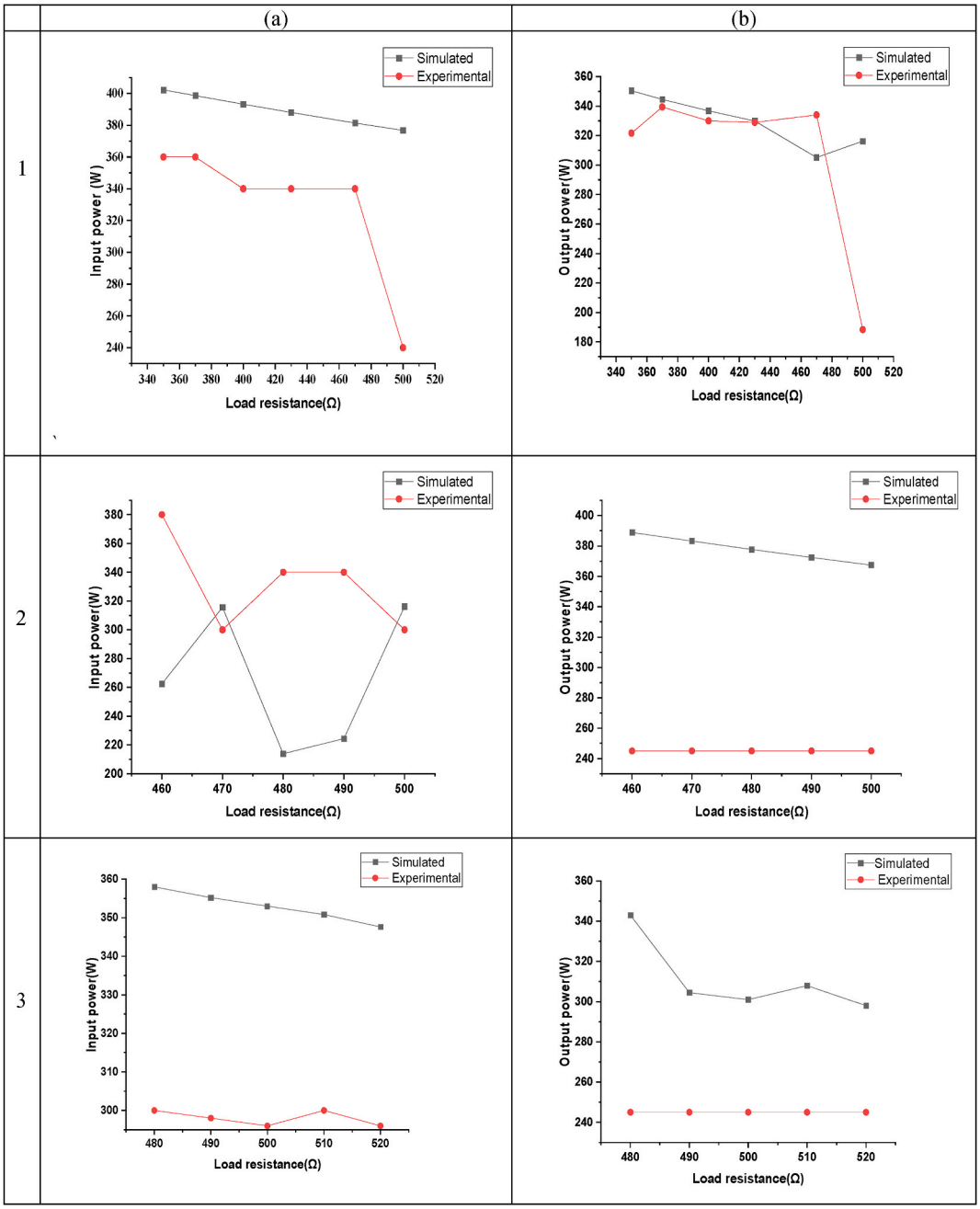


Fig. 16. Comparative load analysis of both simulation and experimental of 1) open loop, 2) PID and 3) Fuzzy logic controllers of the proposed converter.1. a),2. a) and 3. a) are the load resistance Vs input power of the proposed converter. 1. b),2. b) and 3. b) are the load resistance Vs output power of the proposed converter.

controller systems has been compared and depicted. The graphs depict the load resistance versus input power and load resistance versus output power. The load analysis was employed for calculating the input and output power consumption of the converter for various load resistance values. Similarly, Fig. 17 was also implemented with line analysis for converter-based open loop, PID, and fuzzy logic controller systems. The graphs are depicted between input voltage versus input power, as well as input voltage versus output power. Line analysis was employed for calculating the input and output power consumption of the converter for different line voltage values. The PID and fuzzy-based systems consumed the input powers of 286 W and 280 W. The open-loop system consumed the input power of 240 W. The PID and fuzzy-based systems consumed 245 W of output power. The output power consumed by the open-loop system is 188 W.

The proposed converter may be used in battery-operated devices and vehicles by enhancing the performance metrics to the

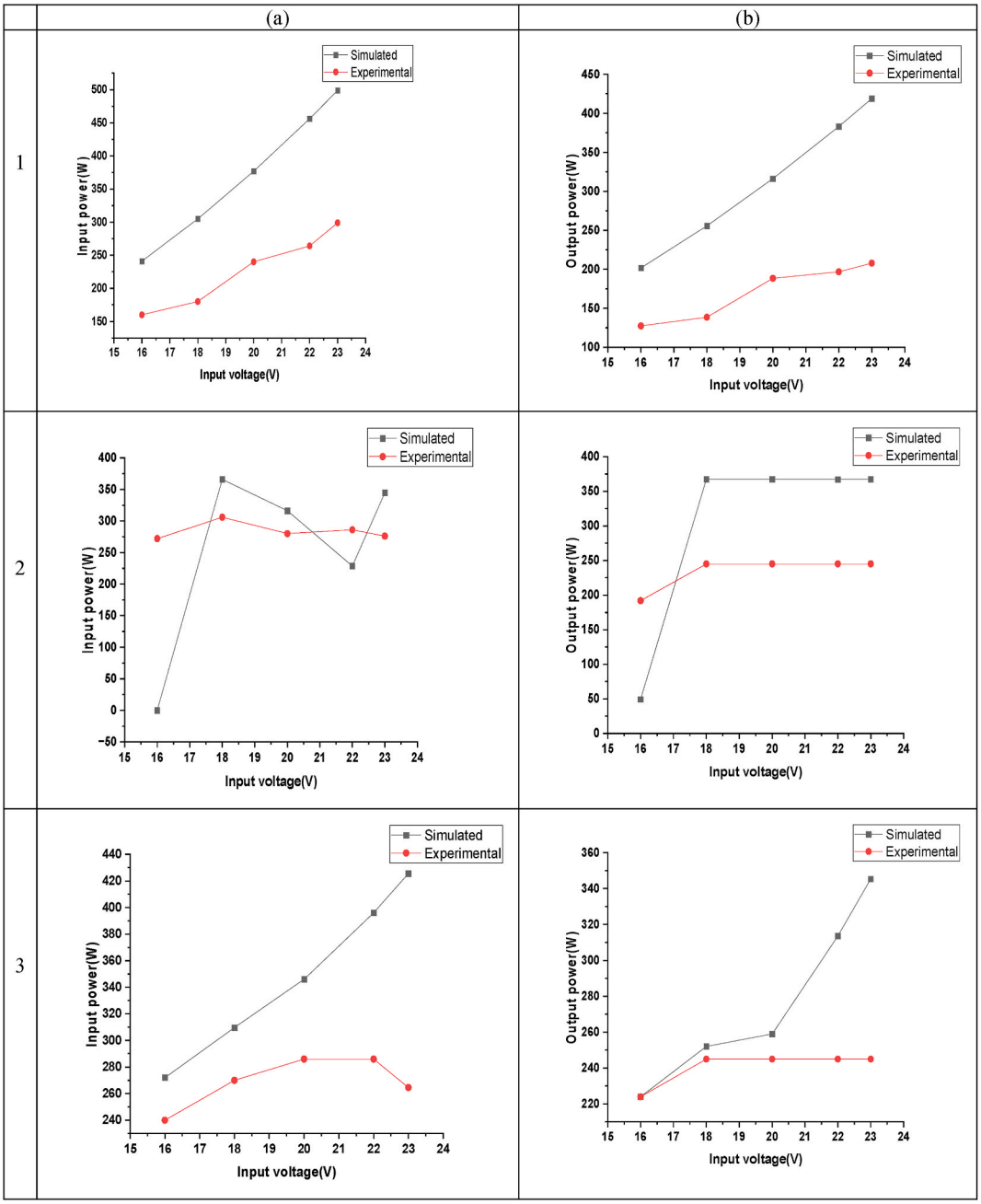


Fig. 17. Comparative line analysis of both simulation and experimental of 1) open loop, 2) PID and 3) Fuzzy logic controllers of the proposed converter.1. a),2. a) and 3. a) are the input voltage Vs input power of the proposed converter. 1. b),2. b) and 3. b) are the input voltage Vs output power of the proposed converter.

required level. This work will be extended, focusing on attaining a fast transient response, good voltage regulation, and high efficiency using the genetic algorithm, particle swarm algorithm [36], and differential evolution algorithm [37] in the proposed converter.

7. Conclusion

The proposed converter was meticulously crafted and simulated using MATLAB to achieve a consistent DC output voltage of 350 V at an output current of 1A, aligning with the V_{max} of the transformer-less inverter application. The line and load analysis of the proposed converter is performed for both the simulation and experimental methods by tuning various line and load values. The line analysis is done by tuning various input voltage values and keeping the load resistance $R_L = 500\ \Omega$ constant. The load analysis is done

by tuning various load resistance values and keeping the input voltage, $V_i = 20V$ constant. In experimental-based line analysis, the PID system achieved an efficiency of 88.76 %, whereas the Fuzzy system achieved an efficiency of 92 %. Also, on comparison of the step-time characteristics of the proposed converter for load resistance $R_L = 500\ \Omega$ and input voltage $V_i = 20V$, the PID system obtained a rise time of 88.781 ms, whereas the Fuzzy system obtained a rise time of 10.624 ms. The rise time of the Fuzzy system is very low compared to the PID system. The PID system attained an undershoot value of 2.000 %, whereas the Fuzzy system attained an undershoot value of 1.99 %.

The PID system achieved an overshoot value of 9.341 %, whereas the Fuzzy system achieved an overshoot value of 0.505 %. The PID system obtained the steady-state error of 0.00043, whereas the Fuzzy system obtained the steady-state error of 0.0584. By comparing the transient parameters of PID and Fuzzy system, the Fuzzy system has the better transient values than the PID system. Performance parameters were fine-tuned and analysed for both PID and Fuzzy logic controllers, with a comprehensive assessment of various time-domain specifications. Measurements of power, output voltage, output current, and transient characteristics were conducted and compared across different controllers on the hardware prototype of the proposed converter implemented with a FPGA-based PID and Fuzzy logic controllers. The simulation and experimental results were compared and analysed for better matching of transient and steady-state values. The results of the performance comparison indicate that, across all operating conditions, the Fuzzy logic controller consistently outperforms the PID controllers. As a result, the authors propose that the boost converter equipped with a Fuzzy logic controller stands out as a champion model to feed transformer-less inverters used in the utility application.

Data availability statement

No data was used for the research described in the article.

CRediT authorship contribution statement

Nethaji G: Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis, Data curation. **Kathirvelan J:** Visualization, Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] P. Dancy and A. P. Chandrakasan, “Ultra Low Power Control Circuits F B R PWM Converters”.
[2] D. Sha, F. You, X. Wang, A high-efficiency current-fed semi-dual-active bridge DC–DC converter for low input voltage applications, IEEE Trans. Ind. Electron. 63 (4) (2016) 2155–2164, https://doi.org/10.1109/TIE.2015.2506625.
[3] Y. Zhang, L. Zhou, M. Sumner, P. Wang, Single-switch, wide voltage-gain range, boost DC–DC converter for fuel cell vehicles, IEEE Trans. Veh. Technol. 67 (1) (2018) 134–145, https://doi.org/10.1109/TVT.2017.2772087.
[4] Y. Zhang, H. Liu, J. Li, M. Sumner, C. Xia, DC–DC boost converter with a wide input range and high voltage gain for fuel cell vehicles, IEEE Trans. Power Electron. 34 (5) (2019) 4100–4111, https://doi.org/10.1109/TPEL.2018.2858443.
[5] L. Guo, J.Y. Hung, R.M. Nelms, Evaluation of DSP-based PID and fuzzy controllers for DC-DC converters, IEEE Trans. Ind. Electron. 56 (6) (2009) 2237–2248, https://doi.org/10.1109/TIE.2009.2016955.
[6] K.W. Seo, H.-H. Choi, Simple fuzzy PID controllers for DC-DC converters, J. Electr. Eng. Technol. 7 (2012), https://doi.org/10.5370/JEET.2012.7.5.724.
[7] L. Guo, J.Y. Hung, R.M. Nelms, Comparative evaluation of linear PID and fuzzy control for a boost converter, IECON Proc. (Industrial Electron. Conf. 2005 (2) (2005) 555–560, https://doi.org/10.1109/IECON.2005.1568965.
[8] C.C. Lee, Fuzzy logic in control systems: fuzzy logic controller. I, IEEE Trans. Syst. Man. Cybern. 20 (2) (1990) 404–418, https://doi.org/10.1109/21.52551.
[9] K.L. Tang, R.J. Mulholland, Comparing fuzzy logic with classical controller designs, IEEE Trans. Syst. Man. Cybern. 17 (6) (1987) 1085–1087, https://doi.org/10.1109/TSMC.1987.6499321.
[10] H. Bevrani, M. Abrishamchian, N. Safari-shad, Nonlinear and linear robust control of switching power converters, in: IEEE Conference on Control Applications - Proceedings, IEEE, 1999, pp. 808–813, https://doi.org/10.1109/cca.1999.807765.
[11] A. Özdemir, Z. Erdem, Double-loop PI controller design of the DC-DC boost converter with a proposed approach for calculation of the controller parameters, Proc. Inst. Mech. Eng. Part I J. Syst. Control Eng. 232 (2) (2018) 137–148, https://doi.org/10.1177/0959651817740006.
[12] N. Swaminathan, Y. Cao, An overview of high-conversion high-voltage DC-DC converters for electrified aviation power distribution system, IEEE Trans. Transp. Electrif. 6 (4) (2020) 1740–1754, https://doi.org/10.1109/TTE.2020.3009152.
[13] J. Kathirvelan, C.D. Viswarani, Design and Analysis of ZCS Based Efficient DC-DC Boost Converter for the Non-conventional Power Units with CV/CC Control, vol. 12, 2012, pp. 23–30.
[14] C. Restrepo, J. Calvente, A. Cid-Pastor, A. El Aroudi, R. Giral, A noninverting buck-boost dc-dc switching converter with high efficiency and wide bandwidth, IEEE Trans. Power Electron. 26 (9) (2011) 2490–2503, https://doi.org/10.1109/TPEL.2011.2108668.
[15] W. Yu, H. Qian, J.S. Lai, Design of high-efficiency bidirectional DCDC converter and high-precision efficiency measurement, IEEE Trans. Power Electron. 25 (3) (2010) 650–658, https://doi.org/10.1109/TPEL.2009.2034265.
[16] X. Wang, F. Tian, I. Batarseh, High efficiency parallel post regulator for wide range input DC-DC converter, IEEE Trans. Power Electron. 23 (2) (Mar. 2008) 852–858, https://doi.org/10.1109/TPEL.2007.915158.
[17] C.-M. Wang, Novel zero-Voltage-transition PWM DC-DC converters, IEEE Trans. Ind. Electron. 53 (1) (2006) 254–262, https://doi.org/10.1109/TIE.2005.862253.
[18] R. Sharma, H. Gao, Low cost high efficiency DC-DC converter for fuel cell powered auxiliary power unit of a heavy vehicle, IEEE Trans. Power Electron. 21 (3) (2006) 587–591, https://doi.org/10.1109/TPEL.2006.872380.
[19] A. Deniz, M. Zeyd, Neuro-fuzzy-SVPWM switched-inductor-capacitor-based boost inverter for grid-tied fuel cell power generators , design and implementation 227 (March) (2024), https://doi.org/10.1016/j.renene.2024.120469.

[20] M. Farhat, O. Barambones, L. Sbita, Ef fi ciency optimization of a DSP-based standalone PV system using a stable single input fuzzy logic controller, Renew. Sustain. Energy Rev. 49 (2015) 907–920, https://doi.org/10.1016/j.rser.2015.04.123.
[21] J.B.L. Fermeiro, J.A.N. Pombo, M.R.A. Calado, S.J.P.S. Mariano, A new controller for DC-DC converters based on particle swarm optimization, Appl. Soft Comput. J. 52 (2017) 418–434, https://doi.org/10.1016/j.asoc.2016.10.025.
[22] M.E. Albira, M.A. Zohdy, Adaptive model predictive control for DC-DC power converters with parameters' uncertainties, IEEE Access 9 (2021) 135121–135131, https://doi.org/10.1109/ACCESS.2021.3113299.
[23] B.P.G.D. Sukumar, Enhanced performance of an asynchronous motor drive with a new modified adaptive neuro-fuzzy inference system- based MPPT controller in interfacing with dSPACE DS-1104, Int. J. Fuzzy Syst. 19 (6) (2017) 1950–1965, https://doi.org/10.1007/s40815-016-0287-5.
[24] S. Seo, H.H. Choi, Digital implementation of fractional order PID-type controller for boost DC – DC converter, IEEE Access 7 (2019) 142652–142662, https://doi.org/10.1109/ACCESS.2019.2945065.
[25] R.S. Maciel, L.C. De Freitas, E. Antô, A. Coelho, Front-end converter with integrated PFC and DC – DC functions for a fuel cell UPS with DSP-based control 30 (8) (2015) 4175–4188, https://doi.org/10.1109/TPEL.2014.2359891.
[26] M. Rezkallah, et al., Hardware implementation of cooperative multitasking control and stability analysis for DC off-grid system, IEEE Trans. Ind. Appl. 58 (3) (2022) 4011–4024, https://doi.org/10.1109/TIA.2022.3151043.
[27] S. Murshid, B. Singh, Implementation of PMSM drive for a solar water, IEEE Trans. Ind. Appl. 55 (5) (2019) 4956–4964, https://doi.org/10.1109/TIA.2019.2924401.
[28] L. Prakash, A.M. Sundaram, S. Jesudaiyan, A simplified time-domain design and implementation of cascaded PI-sliding mode controller for dc – dc converters used in off-grid photovoltaic applications with field test results, Sādhanā 42 (5) (2017) 687–699, https://doi.org/10.1007/s12046-017-0631-y.
[29] K. Cano-pulido, I. Araujo-vargas, Sampled-data model of a two-phase , dual interleaved buck – boost converter with PCM 37 (6) (2022) 6349–6358, https://doi.org/10.1109/TPEL.2021.3125968.
[30] M. Rezkallah, A. Hamadi, A. In, Real-time HIL implementation of sliding mode control for standalone system based on PV array without using dumpload, IEEE Trans. Sustain. Energy 6 (4) (2015) 1389–1398, https://doi.org/10.1109/TSTE.2015.2436333.
[31] S. Member, D. Bortis, J.W. Kolar, ZVS modulation scheme for reduced complexity, IEEE Trans. Power Electron. 33 (5) (2018) 4204–4214, https://doi.org/10.1109/TPEL.2017.2720729.
[32] K. Ogata, Modern Control Engineering [Paperback], 2009.
[33] I. Atacak, O.F. Bay, A type-2 fuzzy logic controller design for buck and boost DC-DC converters, J. Intell. Manuf. 23 (4) (2012) 1023–1034, https://doi.org/10.1007/s10845-010-0388-1.
[34] C. Yanarates, Z. Zhou, Design and cascade PI controller-based robust model reference adaptive control of DC-DC boost converter, IEEE Access 10 (2022) 44909–44922, https://doi.org/10.1109/ACCESS.2022.3169591.
[35] C.-Y. Chan, S. Chincholkar, W. Jiang, A modified fixed current-mode controller for improved performance in quadratic boost converters, IEEE Trans. Circuits Syst. II Express Briefs 67 (10) (2020) 2014–2018, https://doi.org/10.1109/TCSII.2019.2942057.
[36] H.M. Tehrani, A. Frances, R. Asensi, J. Uceda, Blackbox equivalent switching model identification of DC-DC power electronic converters using optimization algorithms, in: 2021 IEEE Fourth International Conference on DC Microgrids (ICDCM), 2021, pp. 1–6, https://doi.org/10.1109/ICDCM50975.2021.9504611.
[37] A. Al-Qallaf, K. El-Sankary, Design of time-mode PI controller for switched-capacitor DC/DC converter using differential evolution algorithm—a design methodology, IEEE Trans. Comput. Des. Integr. Circuits Syst. 41 (12) (2022) 5621–5634, https://doi.org/10.1109/TCAD.2022.3149850.